Sai Adithya Lakku

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Education

Vellore Institute of Technology, Andhra Pradesh 2023 – 2027
Bachelor of Technology in Computer Science and Engineering (AIML) **CGPA: 8.52**

FIITJEE Miyapur 2021 – 2023
Senior Secondary Education **Percentage: 94.2%**

Technical Skills

- **Programming:** Python, Java, SQL, HTML, C, Javascript
- **LLMs & GenAI:** Ollama, OpenRouter, Prompt Engineering, Local Inference, PiperTTS
- **Backend & APIs:** Fast API, Rest API, Flask, PostgreSQL, SQLite, SQLAlchemy
- **DevOps & Tools:** Docker, Git, Linux, Google Colab, Postman

Project Experience

NimbusFS – AI-Enabled File Platform
Python, FastAPI, PostgreSQL, Docker, React, Jinja2, LLM Jun'26 – Present

- Engineered a self-hosted file management platform using FastAPI and PostgreSQL, separating physical disk storage from database metadata for efficient file handling.
- Designed a hierarchical virtual filesystem using PostgreSQL schemas, implementing path-resolution routing and recursive Depth-First Search (DFS) directory deletion.
- Integrated a document summarization pipeline using PyPDF2 text extraction and local Ollama LLM inference (`gemma3:4b`) over an isolated Docker bridge network.
- Built a server-side rendered interface using Jinja2 templates and initiated an API-first React frontend migration to decouple UI components from backend endpoints.
- Orchestrated multi-container deployments (FastAPI application, PostgreSQL database, AI service) using Docker Compose and automated healthchecks.
- Implemented extensible JSON metadata schema storage and enforced filename sanitization controls to prevent path traversal security vulnerabilities.

Local LLM-Based PDF Summarization System
Python, Docker, Ollama, OpenRouter, Discord API, Webhooks Aug'25 – Sep'25

- Built a containerized PDF summarization pipeline using local LLM inference via Ollama.
- Designed Docker-based isolation for reproducible deployments and multi-instance scaling.
- Implemented optional OpenRouter fallback for cloud-based inference.
- Integrated Discord webhooks for asynchronous document submission and result delivery.

Smart AI-Assisted Navigation Stick for Visually Impaired Users
Python, Raspberry Pi 3, OpenCV, YOLOv11n, llamacpp, Piper, Flask,Linux, GPIO Jan'26 – Mar'26

- Engineered a prototype AI-assisted navigation stick using Raspberry Pi 3b+, computer vision, and local LLM-based scene interpretation.
- Orchestrated a client-server pipeline where captured images were processed using YOLOv11n and converted into navigation guidance through a local LLM.
- Integrated Piper text-to-speech on the Raspberry Pi to deliver real-time spoken feedback through earphones.
- Modularized sensor interfacing, networking, inference, and audio workflows for Linux-based embedded deployment.

Certifications

GEN AI using IBM WatsonX *(IBM)* Certification Link Jun'25 – Jul'25